CSE 465: Pattern Recognition and Neural Network

Course Project Proposal

Reliability-Aware V-JEPA for Robust Egocentric Action Anticipation

under Transient Visual Degradation

Working system name: EgoRecover-JEPA

October 14, 2026

Abstract

Egocentric video is naturally affected by rapid head motion, hand–object occlusion, temporary out-of-view events, and other forms of visual degradation. These effects are especially harmful for *action anticipation*, where the most recent frames often contain the strongest evidence about what the camera wearer will do next. This project proposes EgoRecover-JEPA, a reliability-aware latent predictive framework based on V-JEPA 2.1. Instead of attempting to restore corrupted RGB pixels, the method will identify unreliable spatiotemporal observations and use V-JEPA's latent predictor to infer a semantic representation from surrounding temporal context. Observed and predicted representations will then be fused according to estimated reliability before predicting the next verb–noun action. The primary evaluation will use EPIC-KITCHENS-100 under controlled egocentric motion blur and hand–object occlusion with varying severity, duration, and temporal proximity to the anticipation point. The project will compare ordinary augmentation, frame masking/removal, frozen V-JEPA baselines, and the proposed latent recovery strategy. A future extension will investigate missing or noisy gaze as an additional unreliable modality.

1 Background and Motivation

Egocentric video differs from conventional third-person video because the camera moves with the wearer. A recent survey of egocentric vision identifies rapid viewpoint changes, intense motion, occlusion, out-of-view events, and motion blur as major consequences of this highly dynamic setting [1]. The same survey emphasizes that occlusion places strong demands on temporal understanding and memory, because important objects or interactions may disappear from individual frames even though they remain semantically relevant.

Current egocentric systems increasingly use multimodal cues and temporal reasoning to reduce sensitivity to unreliable RGB observations. For example, the surveyed literature includes IMU-assisted action recognition, uncertainty-aware pose estimation, temporal memory for localization, and audio-assisted gaze prediction [1]. However, many systems still process the current visual observation as if its evidence were uniformly reliable.

V-JEPA offers an attractive alternative. Rather than reconstructing pixels, V-JEPA learns by predicting latent visual representations of masked spatiotemporal regions from visible context [2]. V-JEPA 2 extends this idea to large-scale video understanding and reports strong performance on EPIC-KITCHENS-100 action anticipation [3]. V-JEPA 2.1 further improves dense, temporally consistent representations and reports 40.8 Recall@5 on EPIC-KITCHENS action anticipation [4]. These properties motivate our central hypothesis: *when recent egocentric observations become temporarily unreliable, predicting the missing semantic state in latent space may be more useful than attempting to reconstruct the missing pixels.*

2 Problem Statement

Let an egocentric video observation window before an action be

$$X_{t-T:t} = \{F_{t-T}, \ldots, F_t\}, \tag{1}$$

and let the future action be represented as a verb–noun pair

$$A_{t+\Delta} = (V, N). \tag{2}$$

In a clean setting, a model anticipates the future action from all frames. In realistic first-person video, however, the final portion of the observation window may contain corrupted frames

$$\widetilde{F}_{t-k:t} = C(F_{t-k:t}), \tag{3}$$

where C represents motion blur, hand–object occlusion, or both. Because these corrupted frames occur immediately before the anticipation timestamp, they may remove precisely the cues needed to forecast the next interaction.

The project therefore studies the following question:

> **Can a V-JEPA-style latent predictor recover action-relevant semantic information from temporal context when the most recent egocentric observations are unreliable?**

3 Proposed Method: EgoRecover-JEPA

The proposed framework contains four main components: a frozen V-JEPA 2.1 encoder, a reliability estimator, a latent recovery module using the V-JEPA predictor, and an action anticipation head.

3.1 Core Architecture

For a corrupted observation $\widetilde{X}$, the encoder produces observed features

$$z^{\text{obs}} = E(\widetilde{X}). \tag{4}$$

A lightweight reliability head predicts a confidence score for each spatiotemporal token,

$$q_{t,i} = R(z^{\text{obs}}_{t,i}), \qquad q_{t,i} \in [0,1], \tag{5}$$

where low confidence indicates that the token is likely unreliable due to blur or occlusion.

Unreliable tokens are then treated as missing context and recovered with the V-JEPA latent predictor:

$$\hat{z}^{\text{JEPA}}_{t,i} = P(z_{\text{reliable}}, M), \tag{6}$$

where M denotes the locations selected for latent prediction.

The final representation uses reliability-gated fusion:

$$\boxed{z^{*}_{t,i} = q_{t,i} z^{\text{obs}}_{t,i} + (1 - q_{t,i}) \hat{z}^{\text{JEPA}}_{t,i}} \tag{7}$$

so that the model trusts observed features when they are reliable and relies increasingly on predicted semantics when the observation is degraded.

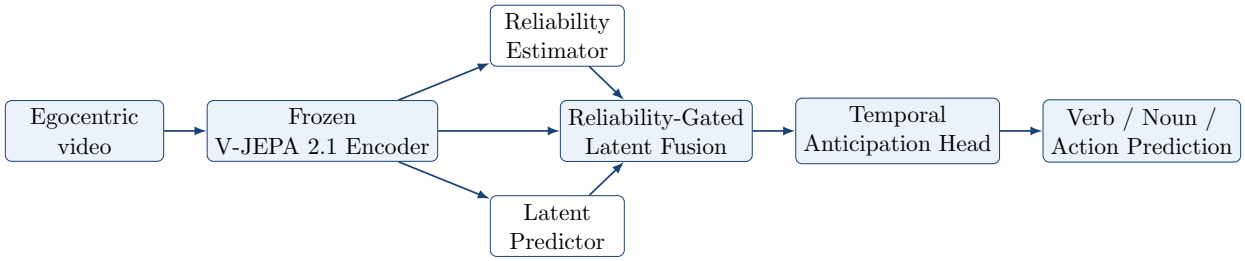


Figure 1: High-level pipeline of the proposed EgoRecover-JEPA framework.

3.2 Training Objective

During training, both a clean clip X and a synthetically degraded version $\widetilde{X} = C(X)$ are available. The clean clip provides a semantic target in latent space:

$$z^{\text{clean}} = E(X). \tag{8}$$

The proposed recovery module is trained to approximate the clean representation from the degraded observation. A practical objective is

$$\mathcal{L} = \mathcal{L}_{\text{action}} + \lambda_1 \mathcal{L}_{\text{latent}} + \lambda_2 \mathcal{L}_{\text{cons}} + \lambda_3 \mathcal{L}_{\text{rel}}, \tag{9}$$

where:

- $\mathcal{L}_{\text{action}}$ supervises verb, noun, and action anticipation;
- $\mathcal{L}_{\text{latent}}$ encourages recovered features to match clean V-JEPA features;
- $\mathcal{L}_{\text{cons}}$ encourages similar predictions for clean and corrupted versions of the same event;
- $\mathcal{L}_{\text{rel}}$ supervises the reliability estimator using the known synthetic corruption mask.

A cosine latent loss may be used:

$$\mathcal{L}_{\text{latent}} = 1 - \cos\left(\hat{z}^{\text{recover}}, \text{sg}(z^{\text{clean}})\right). \tag{10}$$

4 Egocentric Degradation Protocol

The project will focus on corruptions that plausibly arise in wearable-camera video rather than generic image corruption.

Degradation	Egocentric motivation	Planned simulation
Motion blur	Fast head rotation, rapid hand motion, camera shake	Directional or trajectory-dependent blur applied to temporally continuous frames
Hand–object occlusion	The wearer's hand covers the active object during interaction	Hand-shaped or interaction-region masks persisted over consecutive frames
Combined degradation	Real interactions may include blur and occlusion simultaneously	Joint blur + occlusion stress test near the anticipation timestamp

Table 1: Primary degradations considered in the project.

Three factors will be varied independently:

1. **Severity**: weak, medium, and strong degradation;
2. **Duration**: e.g., 0.25 s, 0.5 s, 1.0 s, and 1.5 s;
3. **Temporal proximity**: corruption early in the context window versus immediately before the anticipation point.

This design allows us to study not only *whether* corruption hurts anticipation, but also *when* it becomes most harmful and how long the model can tolerate missing reliable vision.

5 Dataset and Evaluation

5.1 Primary Dataset

We will use **EPIC-KITCHENS-100** as the primary benchmark. EPIC-KITCHENS provides egocentric videos with fine-grained verb–noun interactions and supports an action anticipation benchmark [5]. It is suitable for this project because the videos contain frequent hand–object interaction, natural head motion, and temporally structured actions.

5.2 Evaluation Metrics

The primary task metrics will follow the EPIC-KITCHENS anticipation protocol, including Recall@5 for verbs, nouns, and actions. In addition, the project will report robustness degradation:

$$\Delta R(s) = R_{\text{clean}} - R_{\text{corrupt}}(s), \tag{11}$$

where s denotes corruption severity. The primary objective is not necessarily to maximize clean accuracy, but to reduce the performance loss under degraded observations.

5.3 Baselines and Ablations

The planned comparisons are:

1. Frozen V-JEPA 2.1 on clean video;
2. Frozen V-JEPA 2.1 on corrupted video;
3. Corruption augmentation without latent recovery;
4. Mask/remove unreliable frames or tokens without prediction;
5. A reconstruction-based video representation baseline, subject to available compute;
6. **Proposed latent recovery without reliability gating**;
7. **Full EgoRecover-JEPA with learned reliability gating**;
8. Oracle corruption mask to estimate the upper bound of perfect reliability detection.

The oracle-mask experiment is especially important: if oracle reliability greatly outperforms the learned reliability module, the main bottleneck is corruption detection; otherwise, the limitation lies in latent recovery itself.

6 Research Questions

The project will answer four main questions:

1. How much do realistic motion blur and hand–object occlusion degrade egocentric action anticipation, particularly when they occur immediately before the prediction timestamp?
2. Are V-JEPA latent representations more tolerant to these degradations than conventional reconstruction-based video representations?
3. Can reliability-aware latent prediction recover action-relevant semantics from temporal context when recent visual evidence is unreliable?
4. Does semantic latent recovery outperform simpler strategies such as augmentation, frame removal/masking, or pixel-level restoration?

7 Expected Contributions

The expected contributions of this course project are:

- An egocentric-specific degradation protocol for temporally localized motion blur and hand–object occlusion in action anticipation;
- A lightweight reliability estimator for deciding when observed V-JEPA features should or should not be trusted;
- A latent semantic recovery mechanism that uses V-JEPA prediction instead of pixel reconstruction;
- A robustness analysis across degradation severity, duration, and temporal placement;
- Practical evidence about the limits of latent temporal prediction when recent first-person observations are missing or unreliable.

8 Feasibility and Scope

To keep the project suitable for CSE 465, we will **not train V-JEPA from scratch**. We will begin with a frozen V-JEPA 2.1 backbone and train only the reliability head, latent recovery/fusion module, and anticipation head. If time and compute allow, a later experiment may partially fine-tune selected layers.

The core project is intentionally limited to two primary visual degradations and one primary downstream task. Additional modalities will be treated as future work rather than required components of the first implementation.

9 Tentative Work Plan

Phase	Activities
Week 1	Literature review; reproduce clean V-JEPA 2.1 feature extraction and action anticipation baseline
Week 2	Build motion-blur and hand–object occlusion generation pipeline; verify temporal corruption protocol
Week 3	Measure baseline robustness under severity, duration, and temporal-placement sweeps
Week 4	Implement reliability estimator and corrupted-token masking strategy
Week 5	Implement JEPA latent recovery and reliability-gated fusion
Week 6	Train and tune the proposed model; perform ablations
Week 7	Run final experiments, robustness curves, and error analysis
Week 8	Prepare report, visualizations, presentation, and final code cleanup

Table 2: Tentative eight-week schedule; this can be adjusted to the course calendar.

10 Future Work

The most important extension will be to incorporate **gaze as an additional potentially unreliable modality**. In egocentric applications, gaze can be missing because of eye-tracker dropout, temporary eye occlusion, calibration drift, or rapid head motion. A multimodal extension could encode gaze separately and perform modality-aware reliability estimation:

$$(z_v, z_g) \longrightarrow \text{reliability-aware latent fusion}, \tag{12}$$

where z_v denotes visual features and z_g denotes gaze features. The model could then learn to rely more heavily on video context when gaze is missing and use gaze as an auxiliary cue when RGB becomes visually ambiguous. A future evaluation could use a gaze-annotated egocentric dataset such as EGTEA Gaze+.

Other future directions include incorporating IMU or audio, learning corruption-aware modality routing, extending the framework to short-term object interaction anticipation, and testing whether the same latent recovery strategy transfers to object tracking or egocentric pose estimation.

11 Expected Outcome

A successful result is not defined solely by higher clean-data accuracy. The project will be considered successful if EgoRecover-JEPA maintains approximately the same clean performance as the frozen V-JEPA baseline while showing a consistently smaller performance drop under motion blur, occlusion, and their combination. The guiding principle is:

Do not reconstruct the corrupted pixels; predict the missing semantic state.

References

[1] X. Li, H. Qiu, L. Wang, H. Zhang, C. Qi, L. Han, H. Xiong, and H. Li, "Challenges and Trends in Egocentric Vision: A Survey," *Machine Intelligence Research*, vol. 23, no. 1, pp. 1–33, 2026. DOI: 10.1007/s11633-025-1599-4.

[2] A. Bardes, Q. Garrido, J. Ponce, X. Chen, M. Rabbat, Y. LeCun, M. Assran, and N. Ballas, "Revisiting Feature Prediction for Learning Visual Representations from Video," *arXiv preprint arXiv:2404.08471*, 2024.

[3] M. Assran et al., "V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning," *arXiv preprint arXiv:2506.09985*, 2025.

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